

STATISTICAL ANALYSIS PLAN

Study HTN001

A Phase II, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Drug A 50 mg Once Daily in Subjects with Essential Hypertension

Protocol Number	HTN001
Protocol Version	1.0
SAP Version	1.0
SAP Date	21-Jul-2026
Planned Sample Size	120 subjects
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Document Type	Clinical SAS Portfolio Project

CONFIDENTIAL

This document is prepared for training and portfolio demonstration purposes.

Document Approval

The signatures below indicate review and approval of this Statistical Analysis Plan for the HTN001 mock clinical study.

Role	Name	Signature	Date
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Revision History

Version	Date	Author	Description
1.0	21-Jul-2026	Shanmugam M	Initial version for HTN001 mock study

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1. Introduction

This Statistical Analysis Plan (SAP) describes the planned statistical analyses and presentation of results for Study HTN001. It supplements the protocol by defining analysis populations, derivation rules, statistical summaries, efficacy and safety analyses, and the planned Tables, Listings, and Figures (TLFs).

This SAP is based on the HTN001 protocol, case report form, mock raw database, SDTM datasets, and ADaM dataset specifications prepared for the project. Any deviation from this SAP identified during programming will be documented in the programming notes and final study report.

2. Study Objectives and Endpoints

2.1 Primary Objective

To evaluate the efficacy of Drug A 50 mg once daily compared with placebo in reducing sitting systolic blood pressure after 12 weeks of treatment in subjects with essential hypertension.

2.2 Secondary Objectives

- To evaluate change from baseline in sitting diastolic blood pressure at scheduled post-baseline visits.
- To evaluate the pattern of systolic blood pressure, diastolic blood pressure, and pulse over time.
- To evaluate the safety and tolerability of Drug A using adverse events, treatment exposure, and laboratory assessments.

2.3 Primary Endpoint

Change from baseline in sitting systolic blood pressure (SBP) at Week 12.

2.4 Secondary Endpoints

- Change from baseline in sitting SBP at Week 4 and Week 8.
- Change from baseline in sitting diastolic blood pressure (DBP) at Week 4, Week 8, and Week 12.
- Observed pulse and change from baseline in pulse at scheduled visits.
- Incidence of treatment-emergent adverse events (TEAEs).
- Summary and change from baseline of selected laboratory parameters.

3. Study Design

HTN001 is a Phase II, randomized, double-blind, placebo-controlled, parallel-group study. Approximately 120 subjects are randomized in a 1:1 ratio to receive Drug A 50 mg or placebo once daily for 12 weeks.

Study phase	Phase II
Design	Randomized, double-blind, placebo-controlled, parallel-group
Planned sample size	120 subjects
Treatment arms	Drug A 50 mg QD; Placebo QD
Randomization ratio	1:1
Treatment duration	12 weeks
Planned visits	Screening, Baseline, Week 4, Week 8, Week 12, End of Study; Early Withdrawal if applicable
Primary assessment	Change from baseline in SBP at Week 12

4. Analysis Populations

4.1 Randomized Population

All subjects assigned a randomized treatment. The Randomized Population will be used for subject disposition and randomization summaries.

4.2 Safety Population

All randomized subjects who receive at least one dose of study treatment. In ADSL this population is represented by SAFFL='Y'. Safety analyses will be presented according to actual treatment received (TRT01A).

4.3 Full Analysis Set

All randomized subjects who receive at least one dose of study treatment and have a baseline and at least one post-baseline efficacy assessment. In ADSL this population is represented by FASFL='Y'. Efficacy analyses will be presented according to planned treatment (TRT01P).

4.4 Per-Protocol Set

Subjects in the Full Analysis Set without major protocol deviations considered to affect efficacy evaluation. In ADSL this population is represented by PPSFL='Y'. This population is supportive for the mock project and no separate inferential analysis is planned unless required.

5. General Statistical Considerations

5.1 General Presentation Conventions

- Continuous variables will be summarized using n, mean, standard deviation, median, minimum, and maximum.
- Categorical variables will be summarized using frequency and percentage.
- Percentages will generally use the number of subjects in the relevant treatment and analysis population as denominator.
- Subjects with missing values will be excluded from the denominator for summaries of that variable unless otherwise stated.
- All statistical tests, where performed, will be two-sided with a nominal significance level of 0.05.
- No formal adjustment for multiplicity is planned for this portfolio study.

5.2 Treatment Grouping

Efficacy outputs will use planned treatment (TRT01P). Safety outputs will use actual treatment (TRT01A). A Total column will be included for descriptive summaries where appropriate.

5.3 Baseline Definition

For vital signs and laboratory parameters, baseline is the last non-missing assessment on or before the first dose date (TRTSDT). ADaM variables BASE and BASETYPE will identify the baseline value and method. Change from baseline is calculated as CHG = AVAL - BASE for post-baseline records.

5.4 Visit Windows

For this mock study, nominal visits from SDTM will be mapped directly to analysis visits: Baseline, Week 4, Week 8, and Week 12. No complex visit-window algorithm is required. Unscheduled records will be listed but excluded from visit-based summaries unless selected as baseline.

5.5 Study Day

Study day is derived relative to TRTSDT. For dates on or after TRTSDT, study day equals assessment date minus TRTSDT plus 1. For dates before TRTSDT, study day equals assessment date minus TRTSDT.

6. Subject Disposition

Subject disposition will be summarized by randomized treatment and overall using ADSL. Counts will include screened, randomized, treated, completed, discontinued, and primary reasons for discontinuation when available.

- Randomized subjects: subjects with assigned planned treatment.
- Safety population: subjects with SAFFL=Y.
- Completed subjects: subjects with COMPFL=Y.
- Discontinued subjects: randomized subjects not completing treatment or study.
- Discontinuation reasons: summarized using DCSREAS or corresponding disposition variables when available.

7. Demographic and Baseline Characteristics

Demographic characteristics will be summarized for the Safety Population by actual treatment and overall. Age will be summarized as a continuous variable and by categories 18-40 years, 41-60 years, and greater than 60 years. Sex, race, ethnicity, and country will be summarized categorically.

Baseline vital signs will be summarized from ADVS for SBP, DBP, and pulse using the baseline record identified by ABLFL="Y".

8. Treatment Exposure

Treatment exposure will be summarized for the Safety Population using ADSL and, where needed, SDTM EX. Treatment duration in days will be calculated as $TRTDURD = TRTEDT - TRTSDT + 1$ when both dates are non-missing.

- Number of exposed subjects.
- Planned and actual treatment.
- Dose: Drug A 50 mg or placebo.
- Frequency: once daily (QD).
- Route: oral.
- Treatment duration summarized by n, mean, SD, median, minimum, and maximum.

9. Efficacy Analyses

9.1 Primary Efficacy Analysis

The primary efficacy endpoint is change from baseline in sitting SBP at Week 12. The analysis will use the Full Analysis Set and ADVS records with PARAMCD="SYSBP" and AVISIT="Week 12".

Observed value and change from baseline will be summarized by planned treatment. The treatment difference will be calculated as the Drug A mean change minus the placebo mean change. A two-sample t-test or equivalent linear model may be used as a supportive analysis. Because this is a mock portfolio study, the principal result will be descriptive.

9.2 Secondary Efficacy Analyses

- SBP observed value and change from baseline at Week 4 and Week 8.
- DBP observed value and change from baseline at Week 4, Week 8, and Week 12.
- Pulse observed value and change from baseline at scheduled visits.
- Mean profiles over time for SBP, DBP, and pulse shown in figures.

9.3 Efficacy Data Structure

Variable	Purpose	Expected Values / Rule
PARAMCD	Analysis parameter code	SYSBP, DIABP, PULSE
PARAM	Analysis parameter	Systolic Blood Pressure, Diastolic Blood Pressure, Pulse
AVAL	Analysis value	Numeric result
BASE	Baseline value	Last non-missing pre-dose value
CHG	Change from baseline	AVAL - BASE
AVISIT	Analysis visit	Baseline, Week 4, Week 8, Week 12
AVISITN	Visit number	0, 4, 8, 12
ABLFL	Baseline flag	Y for baseline record

10. Safety Analyses

10.1 Adverse Events

Adverse events will be summarized for the Safety Population by actual treatment. A treatment-emergent adverse event is an adverse event that starts or worsens on or after the first dose date and through the end of the defined safety follow-up period. For the mock project, TEAEFL="Y" will be derived when AESTDT is on or after TRTSDT and no later than TRTEDT plus 7 days, where dates are available.

- Subjects with at least one TEAE.
- Serious TEAEs.
- Severe TEAEs.
- Treatment-related TEAEs.
- TEAEs leading to discontinuation.
- Deaths, if any.
- TEAEs by severity.
- TEAEs by system organ class and preferred term when coded terms are available.

10.2 Laboratory Assessments

Laboratory parameters planned for summary include glucose, alanine aminotransferase, aspartate aminotransferase, creatinine, and hemoglobin, subject to availability in the mock database. Observed values and changes from baseline will be summarized by visit and treatment.

A baseline-to-worst-post-baseline shift table may be produced when categorical normal range classifications are available. Categories will be Low, Normal, and High based on ANRIND or derived reference range comparison.

10.3 Vital Signs Safety Review

Vital signs are also primary and secondary efficacy assessments. Patient-level listings will support review of abnormal or clinically notable values. No formal outlier threshold analysis is planned.

11. Handling of Missing Data

No imputation of missing efficacy or safety values is planned. Analyses will use observed data only. Missing baseline values will result in missing change from baseline. Subjects without a Week 12 assessment will not contribute to the Week 12 observed-case summary.

Partial dates will not be imputed in the current mock project unless needed to determine TEAE status. When TEAE status cannot be determined reliably, the flag will remain missing and the record will be reviewed in a listing.

12. Interim Analysis and Multiplicity

No interim analysis is planned. No formal multiplicity adjustment is planned. The primary endpoint is prespecified; all secondary and supportive analyses are interpreted descriptively.

13. Statistical Software and Programming Standards

- SAS 9.4 or SAS OnDemand for Academics will be used for dataset preparation, statistical summaries, and TLF generation.
- SDTM datasets will follow applicable CDISC SDTM conventions used in the project.
- ADaM datasets will follow applicable CDISC ADaM principles used in the project.
- Tables and listings will generally be produced using PROC REPORT and ODS RTF or ODS PDF.
- Figures will generally be produced using PROC SGPLOT.
- Independent QC will use duplicate programming, targeted checks, PROC COMPARE, record-count checks, and review against approved shells.

14. Planned Tables, Listings, and Figures

The following outputs define the planned TLF package for the HTN001 mock project. Exact layouts will be defined in the separate TLF Shell document.

14.1 Planned Tables

Number	Title	Population	Source
14.1.1	Subject Disposition	Randomized / Safety	ADSL
14.1.2	Demographic and Baseline Characteristics	Safety	ADSL / ADVS
14.1.3	Treatment Exposure	Safety	ADSL / EX
14.2.1	Baseline Vital Signs	FAS	ADVS
14.2.2	Change from Baseline in SBP	FAS	ADVS
14.2.3	Change from Baseline in DBP	FAS	ADVS
14.2.4	Pulse Summary	FAS	ADVS
14.3.1	Overall Summary of Treatment-Emergent Adverse Events	Safety	ADAE
14.3.2	Treatment-Emergent Adverse Events by Severity	Safety	ADAE
14.3.3	Treatment-Emergent Adverse Events by SOC and Preferred Term	Safety	ADAE
14.4.1	Laboratory Summary by Visit	Safety	ADLB
14.4.2	Laboratory Shift Table	Safety	ADLB

14.2 Planned Listings

Number	Title	Population	Source
16.2.1	Subject Demographic Listing	Safety	ADSL
16.2.2	Treatment Exposure Listing	Safety	ADSL / EX
16.2.3	Vital Signs Listing	Safety	ADVS
16.2.4	Adverse Event Listing	Safety	ADAE
16.2.5	Laboratory Results Listing	Safety	ADLB

14.3 Planned Figures

Number	Title	Population	Source
14.2.1	Mean SBP Over Time	FAS	ADVS
14.2.2	Mean DBP Over Time	FAS	ADVS
14.2.3	Mean Pulse Over Time	FAS	ADVS
14.3.1	Treatment-Emergent Adverse Events by Severity	Safety	ADAE
14.4.1	Selected Laboratory Trends Over Time	Safety	ADLB

15. Quality Control and Validation

All key ADaM datasets and TLF outputs will undergo documented QC. Validation will be risk-based and proportional to the mock project scope.

- Confirm expected subject counts by treatment and analysis population.
- Check uniqueness of key variables in ADSL and analysis keys in BDS datasets.
- Verify treatment dates, duration, baseline selection, and change-from-baseline calculations.
- Verify percentages and denominators against source data.
- Independently reproduce critical tables, especially the primary SBP table and overall AE summary.
- Use PROC COMPARE for production and QC datasets or final reporting datasets.
- Review titles, footnotes, pagination, decimal precision, and shell compliance.
- Review SAS logs for errors, warnings, uninitialized variables, and unexpected notes.

16. References

- HTN001 Protocol, Version 1.0, dated 07-Jul-2026.
- HTN001 Case Report Form, Version 1.0.
- HTN001 Variable Specification and mock raw database.
- CDISC SDTM and ADaM implementation principles used for the project.
- SAS 9.4 documentation.

Appendix A. Analysis Variable Requirements

Dataset	Variable(s)	Purpose
ADSL	TRT01P / TRT01PN	Planned treatment and numeric code
ADSL	TRT01A / TRT01AN	Actual treatment and numeric code
ADSL	TRTSDT / TRTEDT	First and last treatment dates
ADSL	TRTDURD	Treatment duration in days
ADSL	SAFFL	Safety Population flag
ADSL	FASFL	Full Analysis Set flag
ADSL	PPSFL	Per-Protocol Set flag
ADSL	COMPFL	Study completion flag
ADSL	AGEGR1 / AGEGR1N	Age group and numeric order
ADVS	PARAM / PARAMCD	Analysis parameter
ADVS	AVAL	Analysis value
ADVS	BASE	Baseline value
ADVS	CHG	Change from baseline
ADVS	AVISIT / AVISITN	Analysis visit and order
ADVS	ABLFL	Baseline record flag
ADAE	TRTEMFL / TEAEFL	Treatment-emergent AE flag
ADAE	AESTDT / AEENDT	Analysis start and end dates
ADAE	AESEV / AESER / AEREL	Severity, seriousness, and relationship
ADLB	AVAL / BASE / CHG	Laboratory analysis value and change
ADLB	ANRIND	Normal range indicator

Appendix B. TLF Output Index

Recommended program naming convention:

- t_14_1_1_disposition.sas
- t_14_1_2_demography.sas
- t_14_1_3_exposure.sas
- t_14_2_2_sbp_change.sas
- t_14_3_1_ae_summary.sas
- l_16_2_1_subjects.sas
- f_14_2_1_sbp_profile.sas

Recommended output formats: RTF for tables and listings; RTF or PDF for figures; optional XLSX export for review only.